

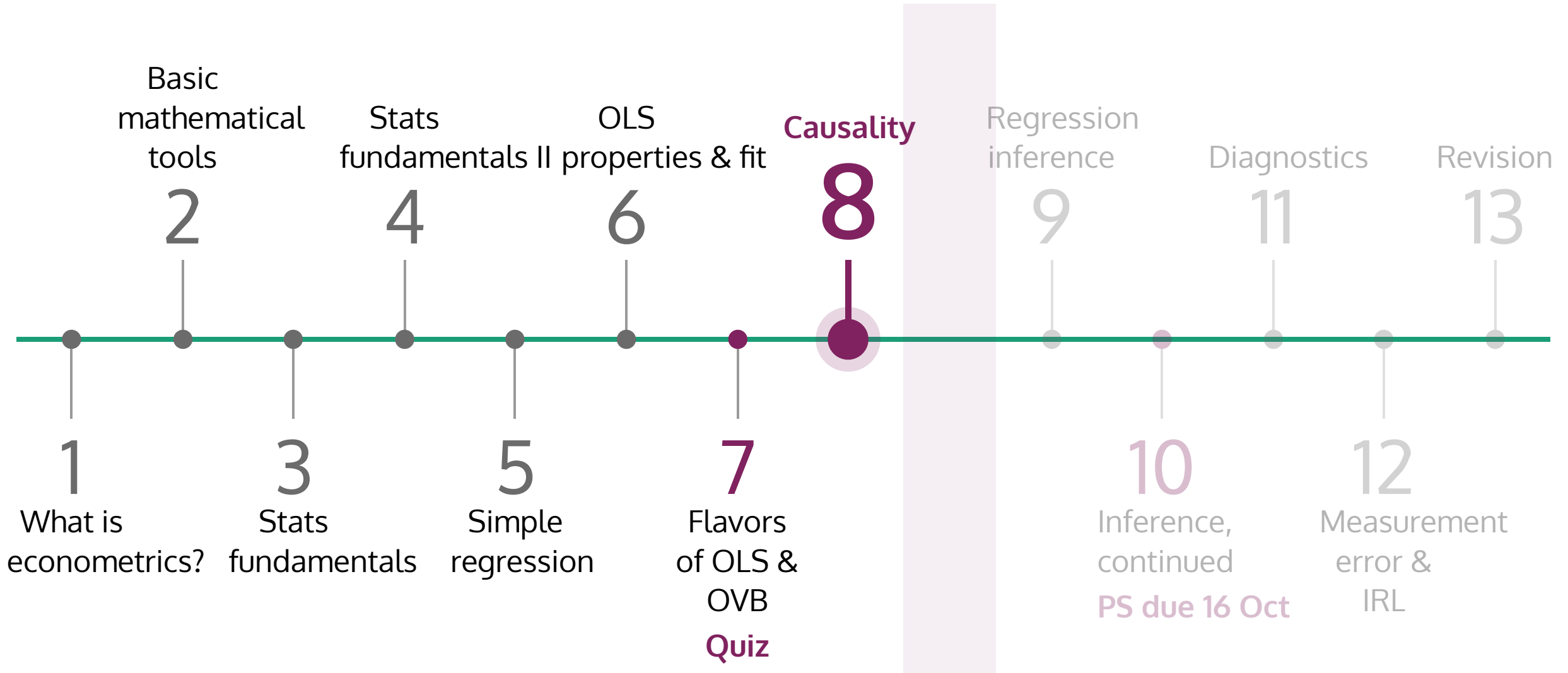


ECON 2041

Introductory Econometrics

More than one  $X$

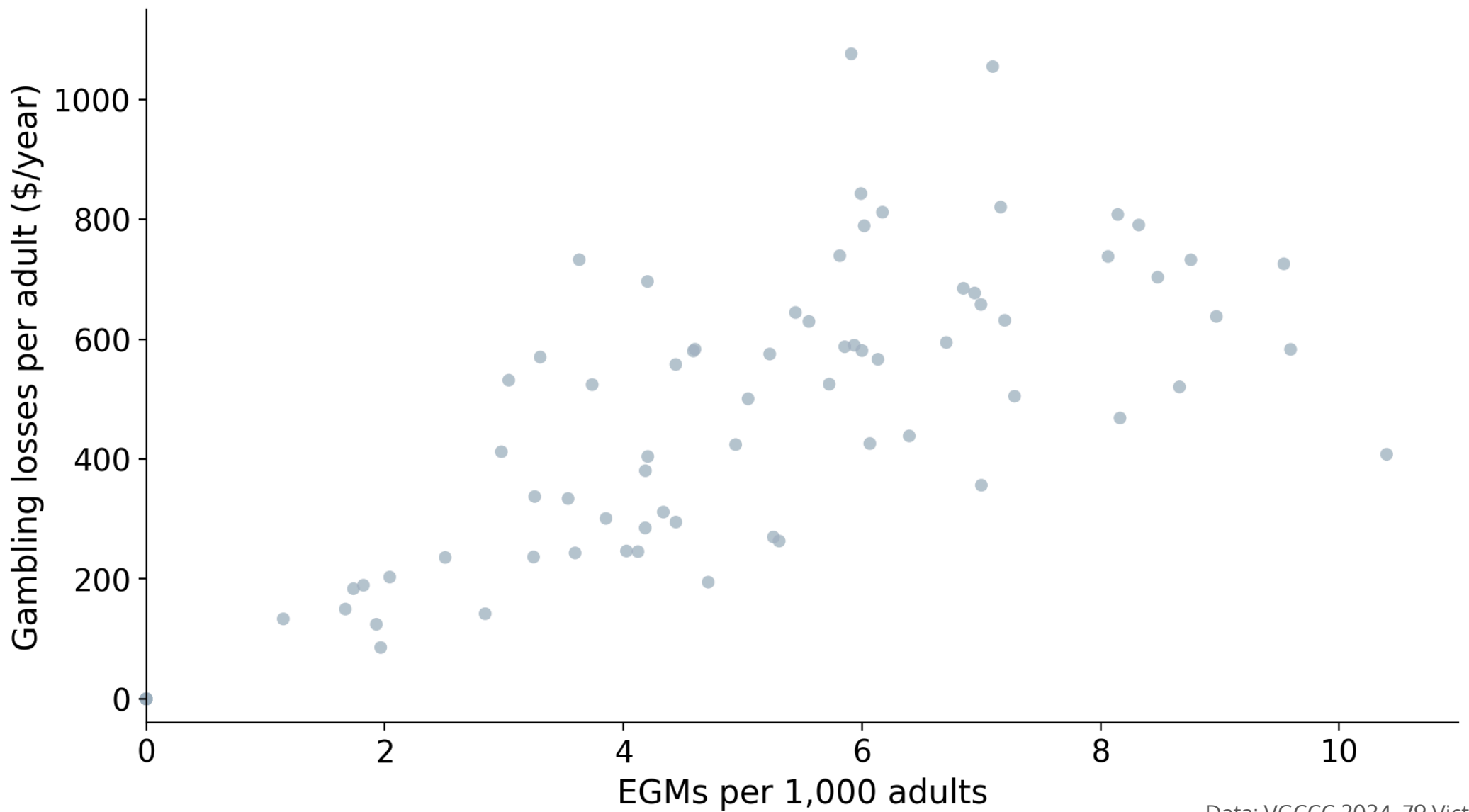
# Plan for the semester



# Plan for today

- Pokies and losses across 79 LGAs
- Unemployment, holding pokie density fixed
- Letting the slopes differ: interactions
- The problem set, in brief
- More than two categories

# Pokie density correlated w/ gambling losses



Data: VGCCC 2024, 79 Victorian LGAs

# What the correlation misses

How many dollars per pokie?

Same method as before

New data

# You can follow along in a notebook

<https://emiliatjernstrom.com/econ2041/lectures/w08>

As usual, it will open directly in Colab

Click **Copy to Drive** first, so you can save your edits

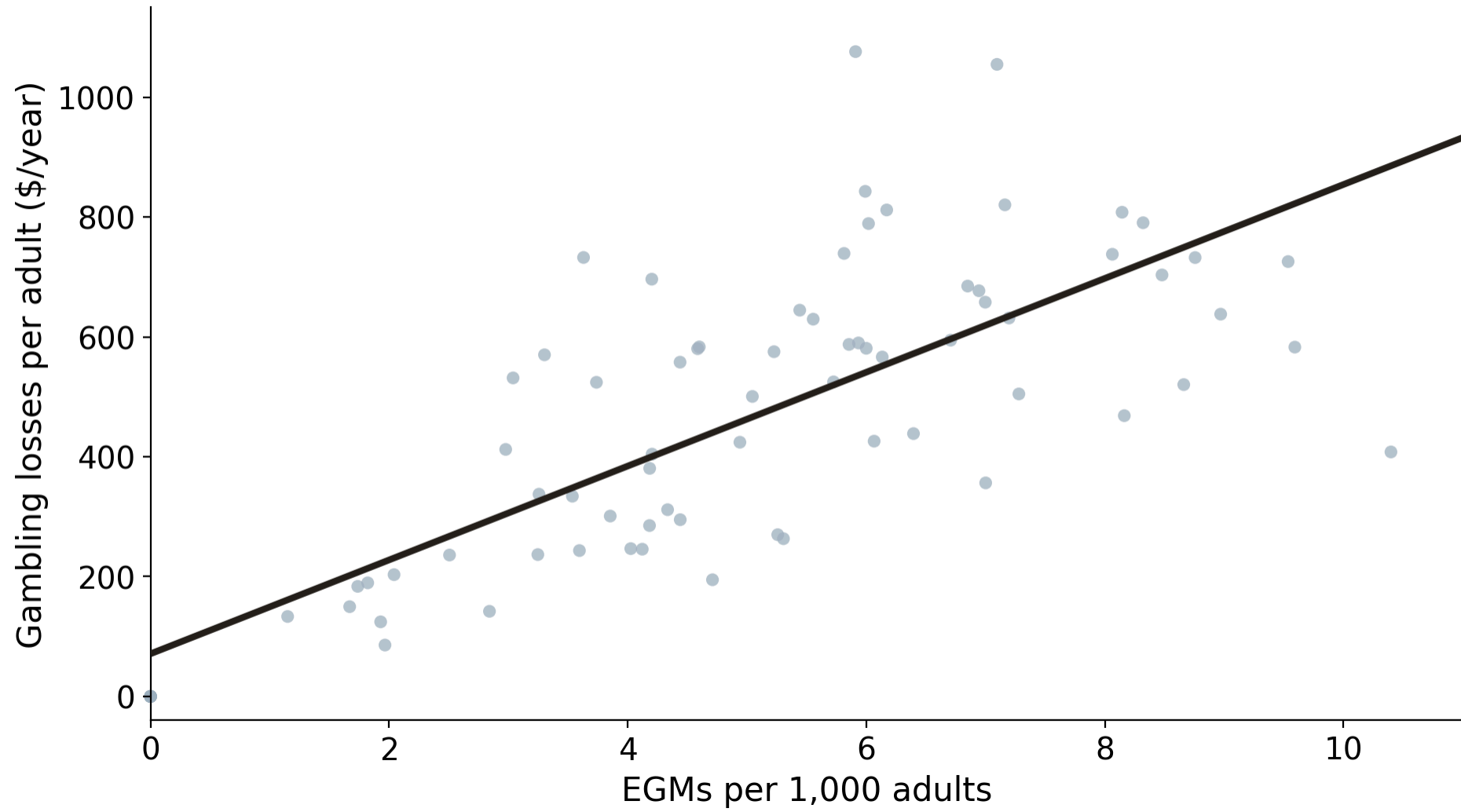
# OLS output: slope & intercept

```
from statsmodels.formula.api import ols
```

```
model = ols("exp_per_adult ~ egm_per_1000", data=pokies).fit()
```

|              | coef    | std err | t      | P> t  | [0.025 | 0.975]  |
|--------------|---------|---------|--------|-------|--------|---------|
| Intercept    | 71.1722 | 38.992  | 1.825  | 0.072 | -6.471 | 148.816 |
| egm_per_1000 | 78.2968 | 7.205   | 10.866 | 0.000 | 63.949 | 92.645  |

# OLS confirms: more pokies, more losses



$$\widehat{\text{losses}} = 71 + 78 \times \text{EGMs per 1,000}$$

Data: VGCCC 2024, 79 Victorian LGAs

# Is unemployment pokie density in disguise?

Our simple regression conditioned on one variable

Take 2 LGAs with the same pokie density, one high-unemployment and one low: does the high one still lose more?

$$E[\text{losses} \mid \text{EGMs, unemployment} = \text{high}]$$

$$E[\text{losses} \mid \text{EGMs, unemployment} = \text{low}]$$

# A dummy variable encodes a category as 0 or 1

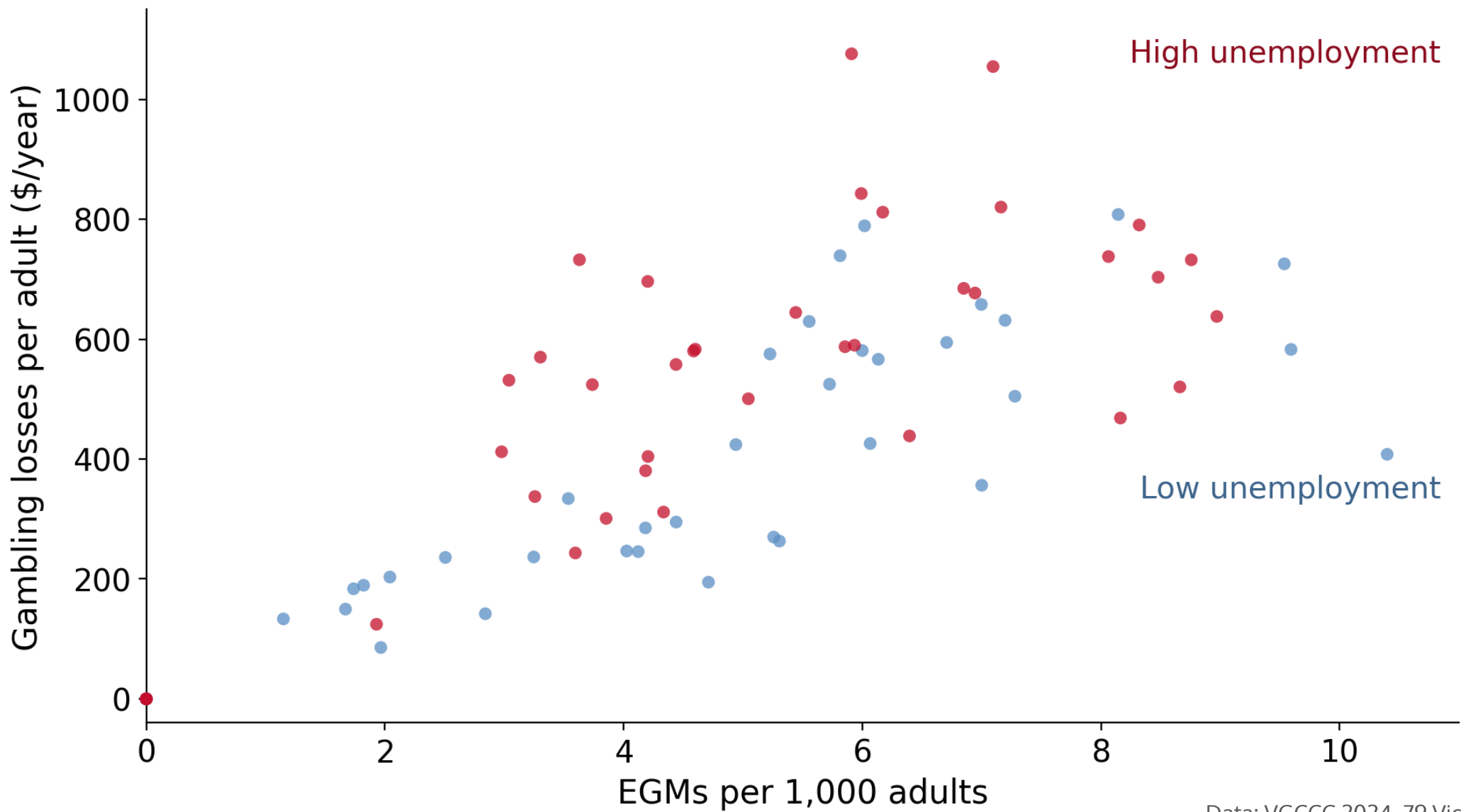
```
median_val = pokies["ue_rate"].median()  
pokies["high_ue"] = (pokies["ue_rate"] > median_val).astype(int)
```

`high_ue` stores the answer to our comparison

`.astype(int)` turns `True/False` into 1 and 0: 1 means above-median unemployment (39 LGAs), 0 means below-median (40 LGAs)

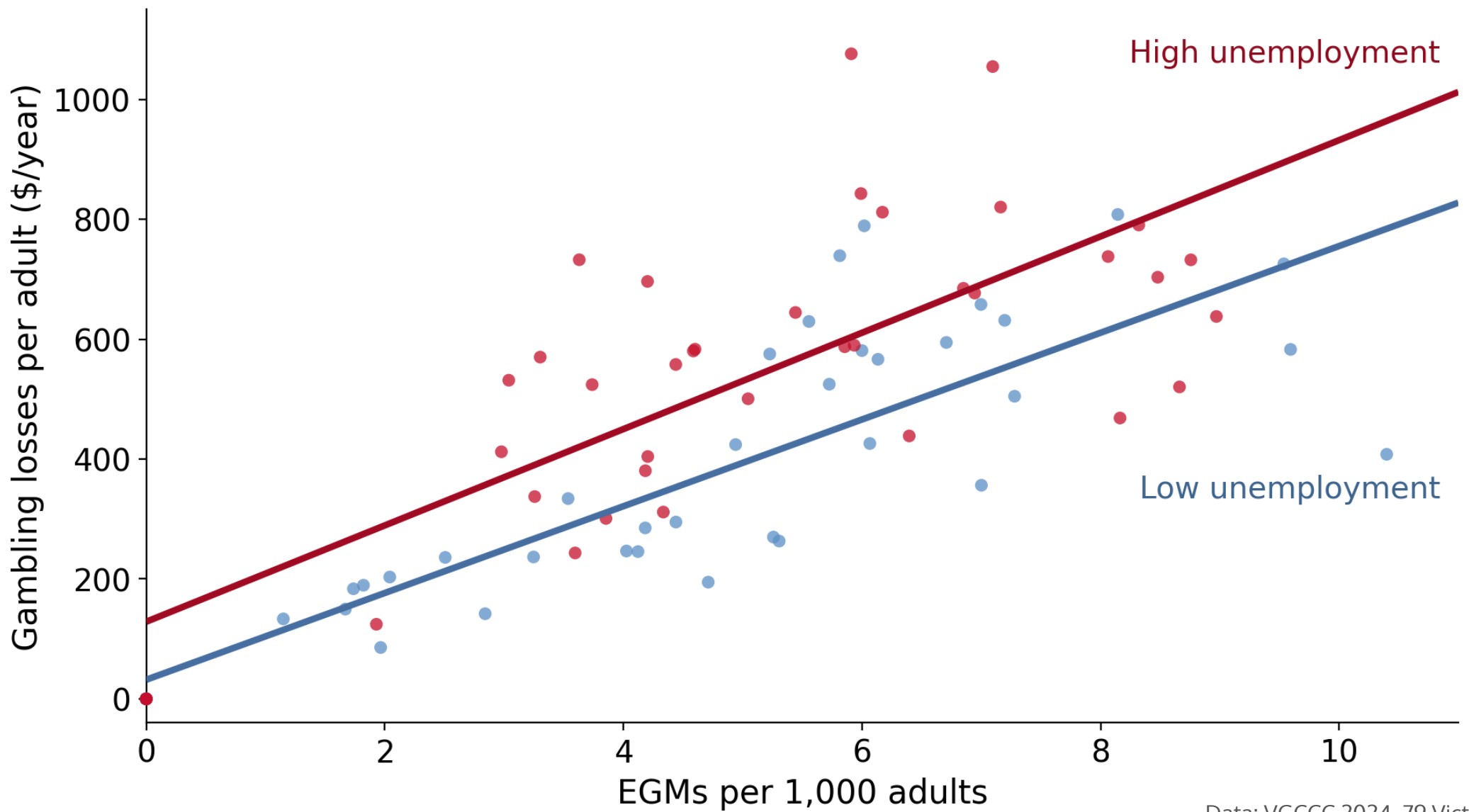
A 0/1 variable is called a **dummy variable**

# Unemployment is not pokie density in disguise



Data: VGCCC 2024, 79 Victorian LGAs

# Unemployment is not pokie density in disguise



Data: VGCCC 2024, 79 Victorian LGAs

# Multiple regression!

Instead of splitting the sample,  
put both variables in one regression  
and use all 79 LGAs

# The dummy shifts the intercept

$$\text{losses}_i = \beta_0 + \beta_1 \text{EGMs}_i + \beta_2 \text{high\_ue}_i + u_i$$

For low-unemployment LGAs ( $\text{high\_ue} = 0$ ): intercept  $\beta_0$

For high-unemployment LGAs ( $\text{high\_ue} = 1$ ): intercept  $\beta_0 + \beta_2$

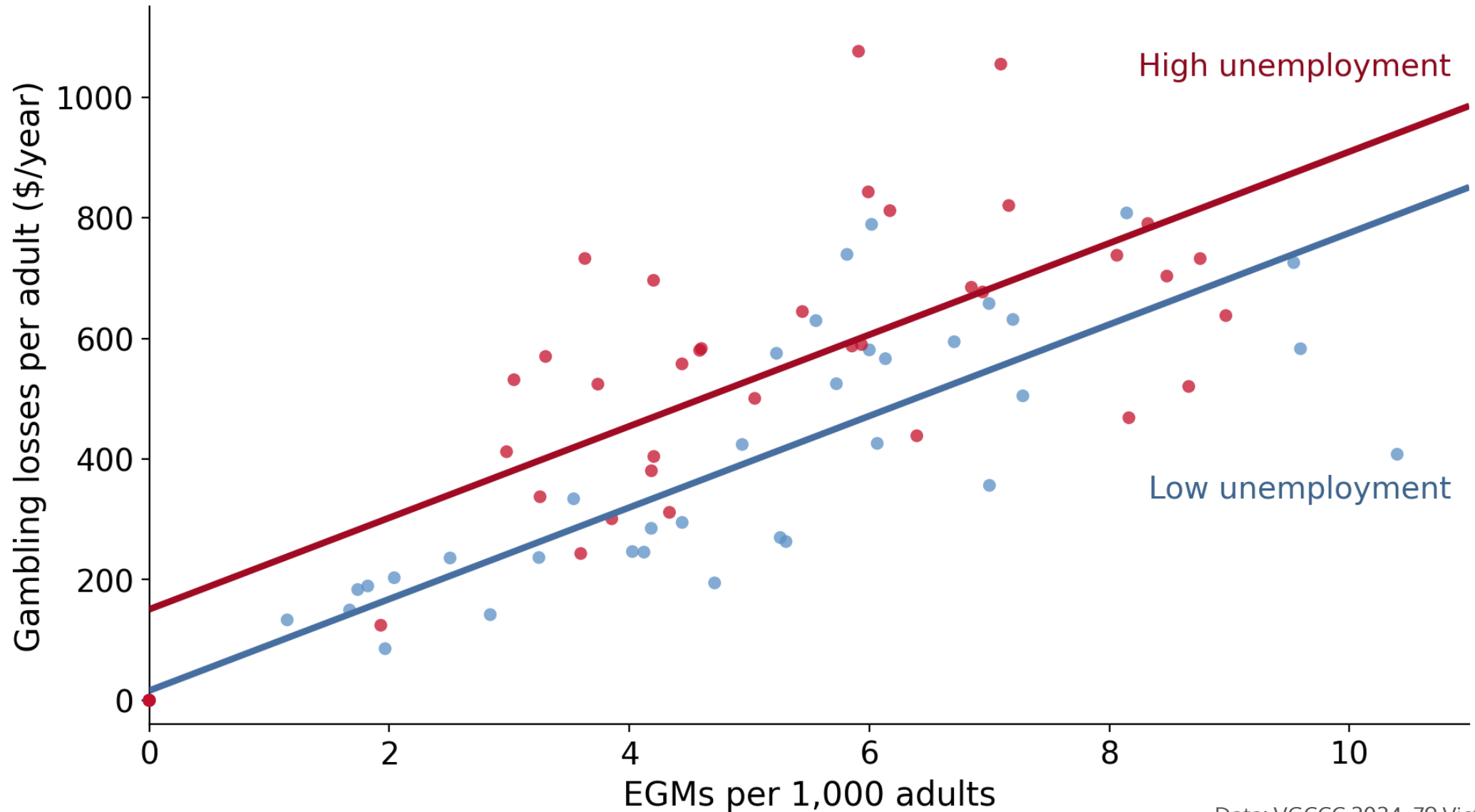
Same slope  $\beta_1$  in both groups: two parallel lines

# Add the high-unemployment dummy

```
ols("exp_per_adult ~ egm_per_1000 + high_ue", data=pokies).fit()
```

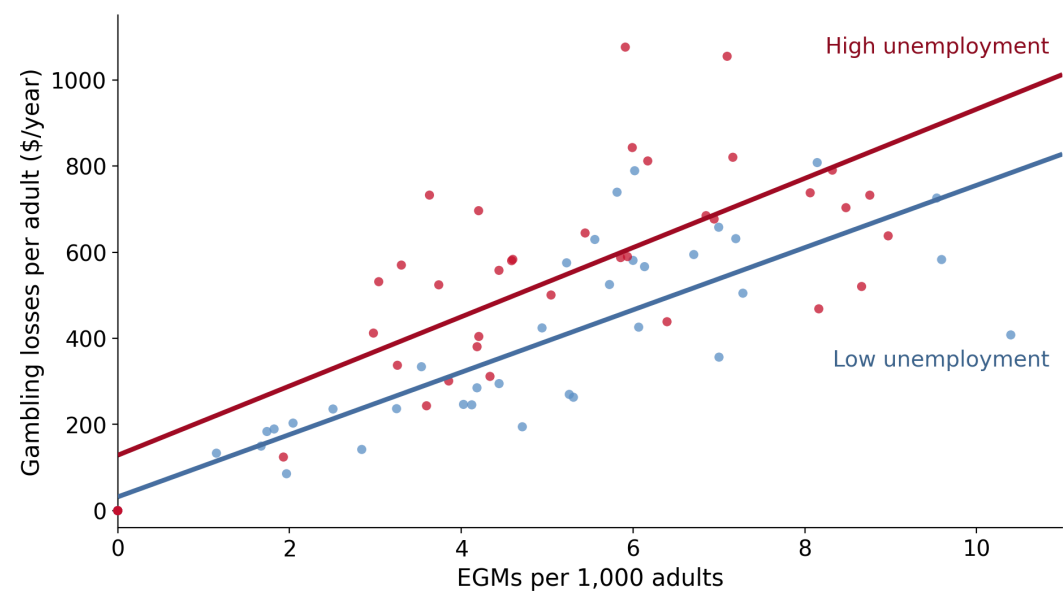
|              | coef     | std err | t      | P> t  | [0.025  | 0.975]  |
|--------------|----------|---------|--------|-------|---------|---------|
| Intercept    | 16.1362  | 38.739  | 0.417  | 0.678 | -61.018 | 93.291  |
| egm_per_1000 | 75.8681  | 6.674   | 11.367 | 0.000 | 62.575  | 89.161  |
| high_ue      | 134.7112 | 35.300  | 3.816  | 0.000 | 64.406  | 205.016 |

# The gap is the same at every pokie density

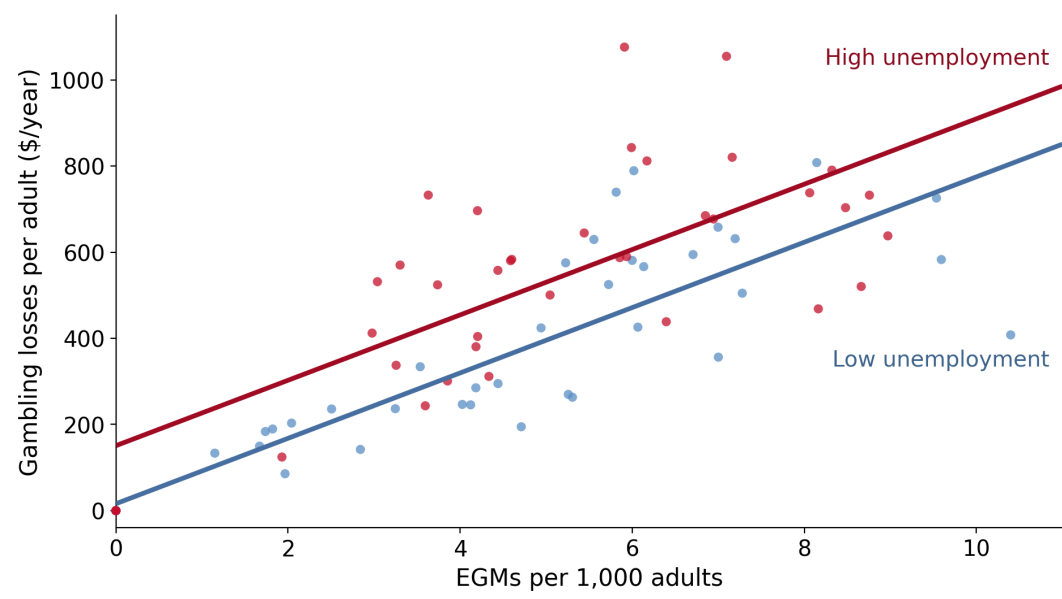


# Separate slopes or one shared slope?

Two regressions, two slopes



One regression, one shared slope



# The second X can be continuous

```
pokies["ue_pct"] = 100 * pokies["ue_rate"]  
ols("exp_per_adult ~ egm_per_1000 + ue_pct", data=pokies).fit()
```

|              | coef      | std err | t      | P> t  | [0.025   | 0.975]  |
|--------------|-----------|---------|--------|-------|----------|---------|
| Intercept    | -115.9316 | 49.520  | -2.341 | 0.022 | -214.559 | -17.304 |
| egm_per_1000 | 72.8291   | 6.329   | 11.507 | 0.000 | 60.224   | 85.434  |
| ue_pct       | 61.1848   | 11.845  | 5.165  | 0.000 | 37.593   | 84.776  |

# Why use one regression instead of splitting?

Uses **all** the data for every coefficient → more precise estimates

Holds the other X fixed: but only the variables we put in the formula

Scales: if we have 3, 4, or 10 X-variables, splitting would not work!

# Letting the slopes differ

One regression

Two intercepts

Two slopes

# Two groups, two slopes, one regression

$$\text{losses}_i = \beta_0 + \beta_1 \text{EGMs}_i + \beta_2 \text{high\_ue}_i + \beta_3 (\text{EGMs}_i \times \text{high\_ue}_i) + u_i$$

Low unemployment ( $\text{high\_ue} = 0$ ): intercept  $\beta_0$ , slope  $\beta_1$

High unemployment ( $\text{high\_ue} = 1$ ): intercept  $\beta_0 + \beta_2$ , slope  $\beta_1 + \beta_3$

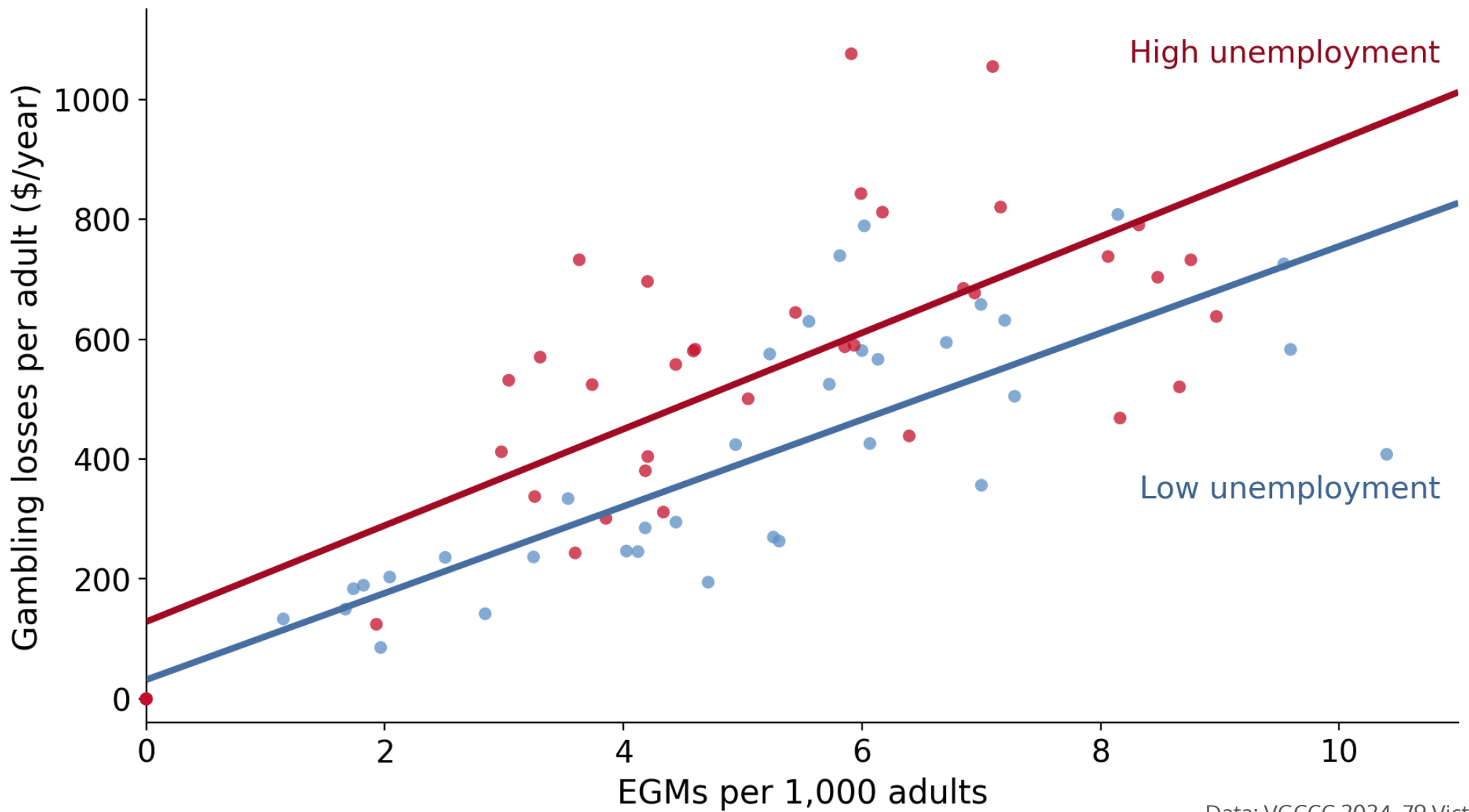
$\beta_3$  is the difference in slopes between the two groups

# In a formula, a colon means interaction

```
ols("exp_per_adult ~ egm_per_1000 + high_ue + egm_per_1000:high_ue",  
    data=pokies).fit()
```

|                      | coef    | std err | t     | P> t  | [0.025  | 0.975]  |
|----------------------|---------|---------|-------|-------|---------|---------|
| Intercept            | 31.9430 | 47.186  | 0.677 | 0.501 | -62.056 | 125.942 |
| egm_per_1000         | 72.3339 | 8.976   | 8.058 | 0.000 | 54.453  | 90.215  |
| high_ue              | 96.7287 | 73.302  | 1.320 | 0.191 | -49.296 | 242.753 |
| egm_per_1000:high_ue | 7.9900  | 13.496  | 0.592 | 0.556 | -18.896 | 34.876  |

# Same fitted values as the split sample



Data: VGCCC 2024, 79 Victorian LGAs

# An asterisk is shorthand for all three terms

```
ols("exp_per_adult ~ egm_per_1000 * high_ue", data=pokies).fit()
```

|                      | coef    | std err | t     | P> t  | [0.025  | 0.975]  |
|----------------------|---------|---------|-------|-------|---------|---------|
| Intercept            | 31.9430 | 47.186  | 0.677 | 0.501 | -62.056 | 125.942 |
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| high_ue              | 96.7287 | 73.302  | 1.320 | 0.191 | -49.296 | 242.753 |
| egm_per_1000:high_ue | 7.9900  | 13.496  | 0.592 | 0.556 | -18.896 | 34.876  |

$a * b$  means  $a + b + a:b$ , so an interaction never enters without its parts

# The problem set, in brief

30% of your grade

Opens Friday 2 October at 9:00, due Friday 16 October at 23:55

- complete it in an iLearn "quiz"
- a short memo
- your notebook

Covers material through week 9

# More than two categories

Low, medium, high

One formula

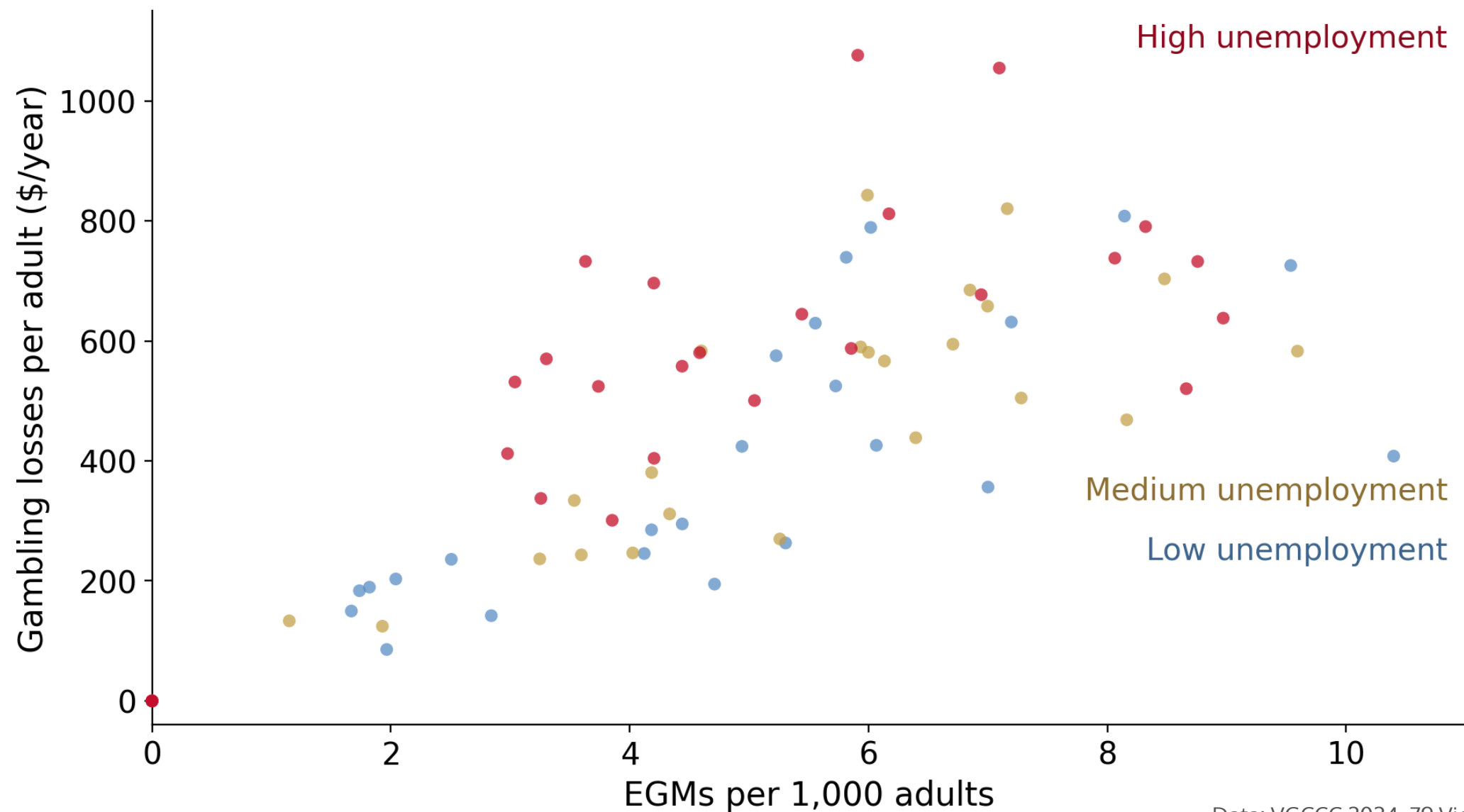
Which numbers?

# Three equal-sized unemployment groups

```
pokies["ue_cat"] = pd.qcut(pokies["ue_pct"], 3,  
                           labels=["low", "medium", "high"])  
pokies["ue_cat"].value_counts()  
  
ue_cat  
low      27  
medium   26  
high     26  
Name: count, dtype: int64
```

`pd.qcut` cuts at the tertiles (2.9% and 4.0%) so the groups are of equal size

# Low & medium unemployment look alike



Data: VGCCC 2024, 79 Victorian LGAs

# Categories have no units, so no single slope

```
# low = 1, medium = 2, high = 3
pokies["ue_code"] = pokies["ue_cat"].cat.codes + 1
ols("exp_per_adult ~ egm_per_1000 + ue_code", data=pokies).fit()
```

|              | coef     | std err | t      | P> t  | [0.025   | 0.975]  |
|--------------|----------|---------|--------|-------|----------|---------|
| Intercept    | -85.6603 | 53.995  | -1.586 | 0.117 | -193.201 | 21.881  |
| egm_per_1000 | 76.4537  | 6.642   | 11.511 | 0.000 | 63.225   | 89.682  |
| ue_code      | 83.2945  | 21.444  | 3.884  | 0.000 | 40.585   | 126.004 |

One coefficient forces the medium-to-high gap to equal the low-to-medium gap

# One dummy per category, minus one

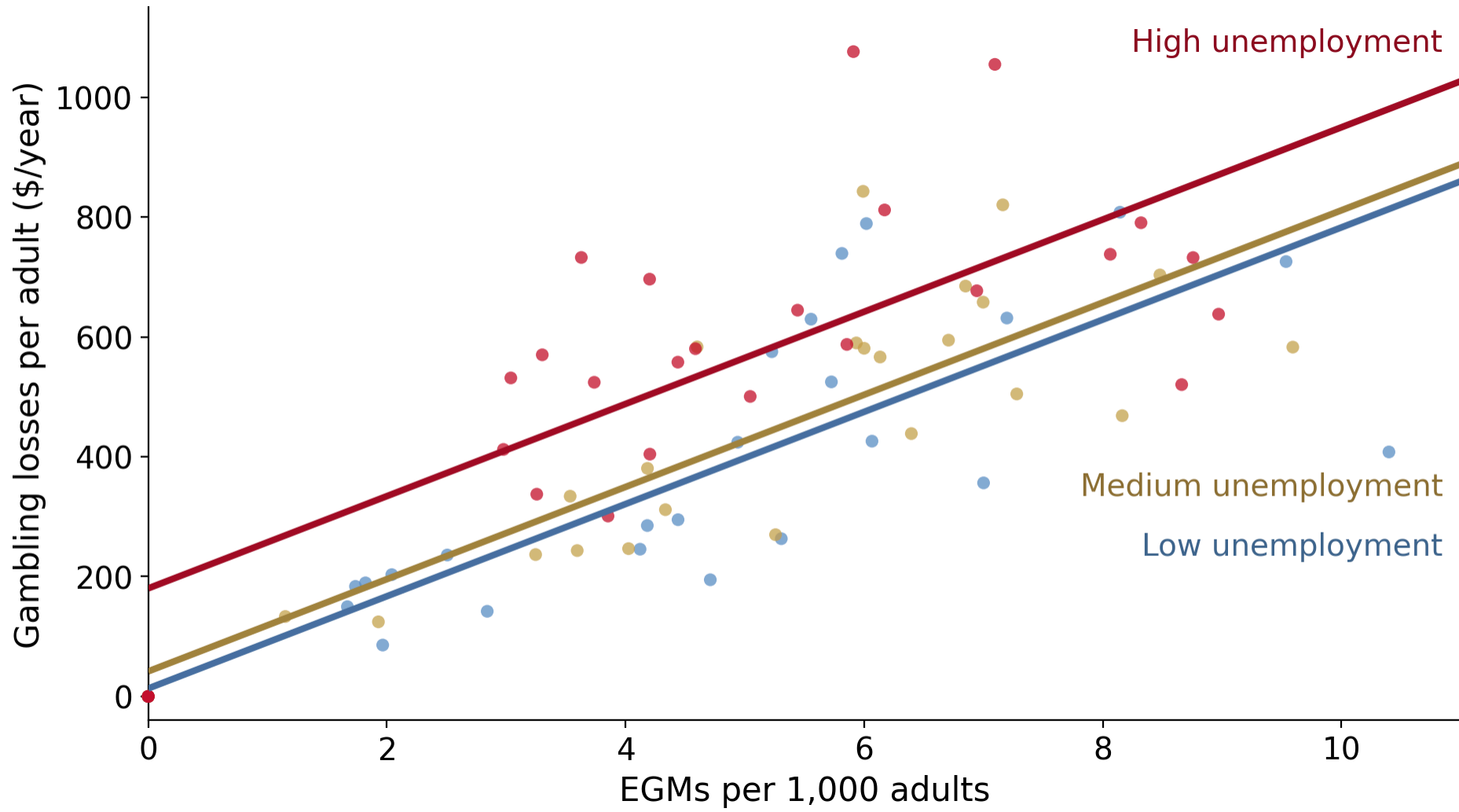
```
ols("exp_per_adult ~ egm_per_1000 + C(ue_cat)", data=pokies).fit()
```

|                     | coef     | std err | t      | P> t  | [0.025  | 0.975]  |
|---------------------|----------|---------|--------|-------|---------|---------|
| Intercept           | 13.2748  | 41.581  | 0.319  | 0.750 | -69.558 | 96.108  |
| C(ue_cat)[T.medium] | 28.5316  | 42.573  | 0.670  | 0.505 | -56.279 | 113.342 |
| C(ue_cat)[T.high]   | 167.0544 | 42.553  | 3.926  | 0.000 | 82.284  | 251.825 |
| egm_per_1000        | 76.9259  | 6.597   | 11.660 | 0.000 | 63.783  | 90.069  |

Low = reference category

i.e., each coefficient is a gap relative to low, at the same pokie density

# Three intercepts, one shared slope



$$\widehat{\text{losses}} = 13 + 77 \times \text{EGMs} + 29 \times \text{medium} + 167 \times \text{high}$$

Data: VGCCC 2024, 79 Victorian LGAs

# The pokie slope is similar in every model

```
summary_col([model, multi, multi_ue, levels, inter], stars=True, float_format="%.2f",
             model_names=["Simple", "Dummy", "Percent", "3 levels", "Interaction"])
```

|                      | Simple             | Dummy                | Percent              | 3 levels             | Interaction        |
|----------------------|--------------------|----------------------|----------------------|----------------------|--------------------|
| Intercept            | 71.17*<br>(38.99)  | 16.14<br>(38.74)     | -115.93**<br>(49.52) | 13.27<br>(41.58)     | 31.94<br>(47.19)   |
| egm_per_1000         | 78.30***<br>(7.21) | 75.87***<br>(6.67)   | 72.83***<br>(6.33)   | 76.93***<br>(6.60)   | 72.33***<br>(8.98) |
| high_ue              |                    | 134.71***<br>(35.30) |                      |                      | 96.73<br>(73.30)   |
| ue_pct               |                    |                      | 61.18***<br>(11.85)  |                      |                    |
| C(ue_cat)[T.medium]  |                    |                      |                      | 28.53<br>(42.57)     |                    |
| C(ue_cat)[T.high]    |                    |                      |                      | 167.05***<br>(42.55) |                    |
| egm_per_1000:high_ue |                    |                      |                      |                      | 7.99<br>(13.50)    |
| R-squared            | 0.61               | 0.67                 | 0.71                 | 0.68                 | 0.67               |
| R-squared Adj.       | 0.60               | 0.66                 | 0.70                 | 0.67                 | 0.66               |

# Holding fixed only works for what we measured

The slope on pokie density: 78 alone, 76 with the dummy, 73 with the percentage

Each extra variable holds fixed one more thing, and only that one thing

Remoteness, venue rules: unmeasured, so still in the slope

Today's estimates aren't causal